10-bounded-degree — results

Item 10 of the pair-46 factor-base campaign (2026-10-15): bounded-degree algebraic factor bases over prime fields, the repaired form of `IDEA-20261009-a7733f` / design audit `ALGEBRAIC-FACTORBASE-DIMENSION-20261010`. Protocol: `PROTOCOL.md`, frozen before data, sha256 `e2b18e84a8fcf524215dba216814ec4a85f5d80347cf308810d23587e2bcd3a9` (`PROTOCOL.sha256`, also embedded in `raw.json`).

Verdict: exact obstruction, confirmed by a finite sanity check. Every bounded-degree family (degree 2..8 polynomial and rational x-images, power and Dickson maps) has `|F|/p >= 0.167` and exponent `log|F|/log p >= 0.839` at p ~ 2^16..2^24; the extension-field positive control gives exponent 0.499..0.505. Random families sit on the Chebotarev value c_d to within +-0.003. Zero-dimensional conditions give `|F| <= 4`. This closes only the tested families; it is not a theorem about all subsets of E(F_p).

1. Design-audit repair (what changed before data)

Audit item	Repair
`p in {2^16,2^20,2^24}` are not primes	p_k = smallest prime `> 2^k`, `= 3 mod 4`: **65539 (17 bits), 1048583 (21 bits), 16777259 (25 bits)**; deterministic MR (12 bases)
Extension control undefined	q = 263, 1031, 4099 (`= 3 mod 4`), `F_{q^2}=F_q[i]/(i^2+1)`, irreducibility checked; `F = {x in F_q}`
"factor base" not a subset of E	every F is a set of affine points of E(F_p), union counted once, O excluded
Hasse-Weil hypotheses; "genus p^0.4"	exact bound X2 stated for squarefree `h_f` (all 168 algebraic rows were squarefree; genus computed exactly by Yun decomposition); threshold corrected (X3)
Representation size not modelled	dense coefficient bits recorded per row; membership cost counted in F_p mults on the 53-bit pair-46 curve
Relation/LA costs	cut (preregistered): X2 already gives `#F >> sqrt(p)`

2. Exact statements (proved, not measured)

• **X1.** A degree-d map `P^1 -> P^1` has `|image(F_p)| >= (p+1)/d`.

• **X2.** `|F_f| >= LB_f = (p + 1 - 2 g_f sqrt(p) - 3 - 2 deg Q)/d`, where `C_f: y^2 = h_f(t)` maps onto `F_f` with fibers `<= d`. Genus: `g = floor((3d-1)/2)` for polynomials, `2d-1` for generic rational maps (measured exactly per row, matches).

• **X3 (corrects the proposal's p^0.4).** With `g <= 2d-1`: `|F| <= sqrt(p)` forces `d >= sqrt(p)/5 (1-o(1))`; `|F| < p^0.9` forces `d >~ p^0.1/5`. For any cover `C -> E` of degree d and genus g, `|F| >= (p+1-2g sqrt p)/d`, so sub-`p^0.9` needs `d >~ p^0.1` or `g >~ sqrt(p)/2`. For pair 46 (`sqrt p = 7.71e7`): `|F| <= sqrt(p)` needs `d >= 1.54e7`, dense size `>= 8.2e8` bits.

• **X4.** `A(x) + yB(x) = 0` (deg A <= d, deg B <= d-2) has `<= 2d` zeros on E.

3. Controls

Control	Result
Fields: MR prime, `p = 3 mod 4`, Hasse, `[#E]R = O` for 4 points	pass at all 3 primes (#E = 65736, 1048896, 16776014)
Negative (i) identity map d=1	`#F = #E - 1` exactly at all 3 primes
Negative (ii) RF random function (9 rows)	rho = 0.6327, 95% CI [0.6319, 0.6334] (1 - 1/e = 0.6321)
Positive (i) EXT x in F_q over F_{q^2}	#F = 278, 1058, 4006 for q = 263, 1031, 4099; exponent 0.505, 0.502, 0.499: **valid**
Positive (ii) PW image = `(p-1)/gcd(d,p-1)+1`; DK permutation when `gcd(d,p^2-1)=1`	all 42 rows exact
X2 lower bound holds (implementation check)	168/168 rows
X4 upper bound holds	63/63 ZD rows (max #F = 4)

4. Measured: random families vs Chebotarev value c_d

n = 9 per cell (3 primes x 3 seeds), 95% t-interval. `Delta = rho - c_d`.

d	c_d	RP rho [95% CI]	RP Delta [95% CI]	RR rho [95% CI]	RR Delta [95% CI]	genus RP / RR
2	0.5000	0.5003 [0.4999, 0.5007]	[-0.0001, 0.0007]	0.5006 [0.4994, 0.5017]	[-0.0006, 0.0017]	2 / 3
3	0.6667	0.6678 [0.6664, 0.6691]	[-0.0003, 0.0025]	0.6669 [0.6665, 0.6672]	[-0.0002, 0.0006]	4 / 5
4	0.6250	0.6253 [0.6247, 0.6260]	[-0.0003, 0.0010]	0.6257 [0.6245, 0.6268]	[-0.0005, 0.0018]	5 / 7
5	0.6333	0.6334 [0.6324, 0.6345]	[-0.0010, 0.0012]	0.6344 [0.6329, 0.6358]	[-0.0004, 0.0025]	7 / 9
6	0.6319	0.6331 [0.6314, 0.6348]	[-0.0006, 0.0029]	0.6325 [0.6312, 0.6338]	[-0.0007, 0.0018]	8 / 11
7	0.6321	0.6324 [0.6316, 0.6331]	[-0.0006, 0.0010]	0.6334 [0.6324, 0.6345]	[0.0002, 0.0023]	10 / 13
8	0.6321	0.6334 [0.6316, 0.6352]	[-0.0005, 0.0031]	0.6328 [0.6314, 0.6343]	[-0.0008, 0.0022]	11 / 15

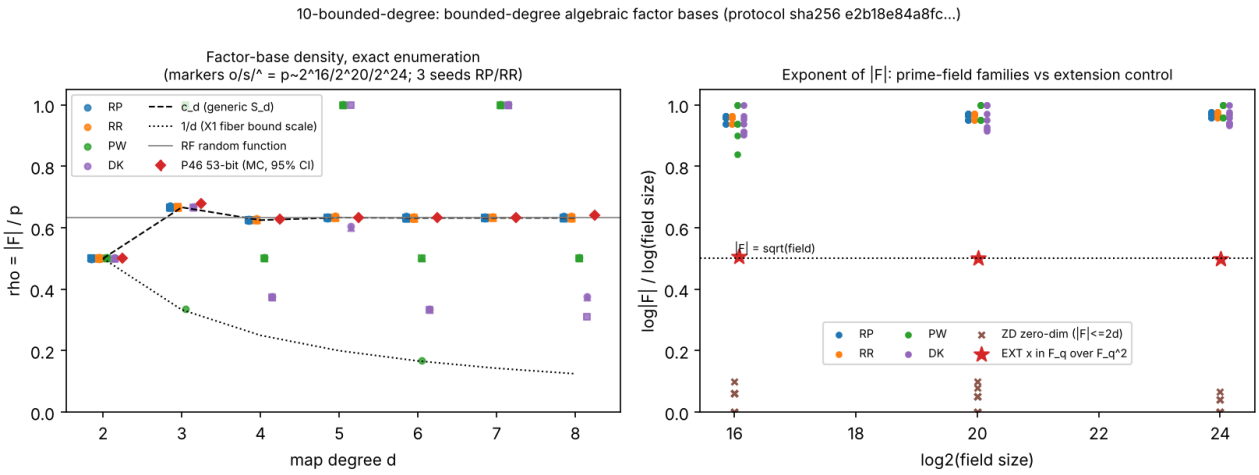
13 of 14 Delta intervals contain 0; RR d=7 is +0.0013 [0.0002, 0.0023]
(finite-p fluctuation of order `p^-1/2`; no correction for 14 comparisons).
Special maps reach lower density only through the fiber structure X1: the
minimum over all algebraic rows is PW `t^6` at p = 65539 (`gcd(6,p-1) = 6`),
rho = 0.1673, `|F|` = 10967, exponent 0.839 — X1's `1/d` scale, attained.

5. Pair 46 (53-bit crater curve), Monte Carlo

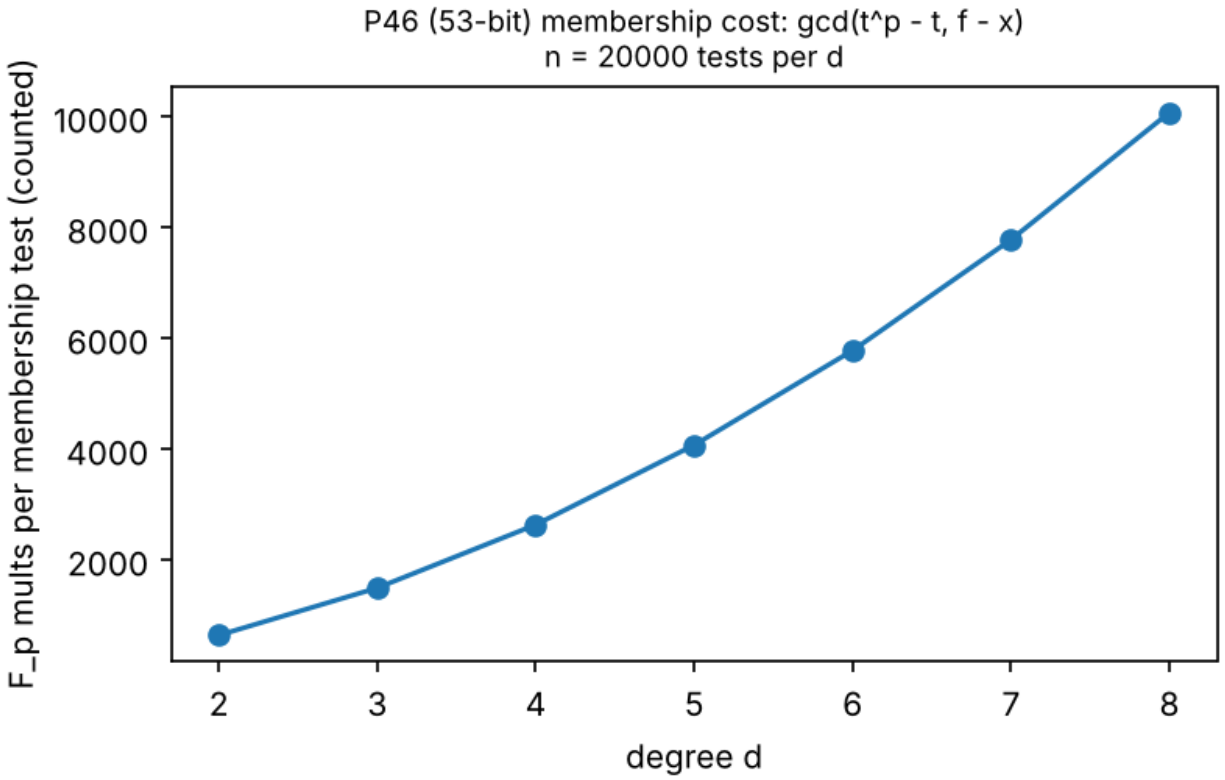
`p = 5947698030651343`, crater `a, b` as in the campaign brief; one random
monic polynomial per d (coefficients in `raw.json`), n = 20000 uniform x,
95% normal interval. Membership = `gcd(t^p - t, f(t) - x) != 1`; cost counted
in F_p multiplications in the instrument.

d	rho_hat [95% CI]	c_d	F_p mults / membership	dense size (bits)
2	0.5012 [0.4892, 0.5132]	0.5000	639	159
3	0.6785 [0.6654, 0.6916]	0.6667	1493	212
4	0.6276 [0.6147, 0.6405]	0.6250	2634	265
5	0.6323 [0.6194, 0.6452]	0.6333	4072	318
6	0.6322 [0.6193, 0.6451]	0.6319	5779	371
7	0.6336 [0.6207, 0.6465]	0.6321	7780	424
8	0.6411 [0.6282, 0.6540]	0.6321	10075	477

Membership is cheap (about `160 d^2` mults, i.e. `O(d^2 log p)`); the
obstruction is size, not membership: every interval contains c_d, so `|F|`
is about `0.63 p = 3.8e15` points against `sqrt(p) = 7.7e7`.



Factor-base density rho vs degree (left) and exponent log|F|/log(field) vs field size including the extension positive control (right); raw.json, protocol e2b18e84



Pair-46 membership cost per test vs degree, n = 20000 per d

6. Exact vs empirical

• Exact: X1-X4; all row counts (`|image|`, `|F|`, #E, genus) are exact enumerations, not estimates.

• Empirical: agreement with c_d (E1) and the pair-46 densities (E2, Monte Carlo).

• Scope: x-images of degree <= 8 polynomial and rational maps, power and Dickson maps, zero-dimensional `A + yB` conditions, at three toy primes plus pair 46 sampled. Not tested: other covers `C -> E` (their bound is X3, exact), compact high-genus descriptions (not excluded by any finite run).

7. Deviations and cuts

• Nothing cut from the preregistered design; wall clock 87.6 s on one core (`taskset -c 3`).

• RF was run once per prime (3 seeds), not once per d, as it has no degree.

• EXT curves were checked by nonsingularity and Hasse only; the `[#E]R = 0` check was run only for prime-field curves.

• P46 polynomial seed is derived from 1825564914 mixed with d (see `src/main.rs`, `p46`); the coefficients are recorded.

• Relation generation and linear algebra not run (preregistered cut).

8. Files and reproduce

`src/main.rs` (Rust instrument; all counts), `raw.json` (instrument output),

`analyze.py` (summary statistics and plots of recorded values), `results.json`,

`fig_density.{svg,png}`, `fig_membership.{svg,png}`, `make_pdf.py` (layout),

`report.pdf`, `run.log`.

```
cd research/pair46_factor_bases_20261015/10-bounded-degree
cargo build --release && taskset -c 3 ./target/release/bounded_degree 20000 2> run.log
python3 analyze.py . && python3 make_pdf.py .
```